Detecting Hallucinations Through Compositional Inconsistency: A Categorical Approach to Budgeted Verification

Abstract

Large language models can correctly recall individual facts while failing to combine them into a correct answer. This project investigates whether failures of compositional consistency provide a useful signal for detecting factual hallucinations and selecting claims for verification. We represent a multi-step task as a graph of typed operations and use category theory to specify when different computational paths should produce equivalent answers. We then measure disagreement between a model's direct execution of a composite task and its execution of the constituent steps, without accessing ground-truth answers during detection.

Using multi-hop question-answering datasets with independently checked intermediate facts, we will test whether these composition failures predict factual errors beyond confidence, reasoning depth, task difficulty, and ordinary answer disagreement. We will subsequently evaluate whether this signal improves verification decisions under a fixed total budget, including the cost of diagnostic model calls. The proposed contribution is an operational evaluation of composition failures as a detection and prioritization signal. Because models can be consistently wrong, the project treats categorical consistency as a testable indicator of reliability rather than a guarantee of truth.

1 Research Goal and Motivation

The central research question is:

> Can failures to preserve task composition help identify factual hallucinations and determine which claims are most valuable to verify?

The motivation is that an answer's reliability depends on more than the correctness of its individual facts. A model must also preserve the relationships between those facts as it combines them.

For example, answering "In which country is the headquarters of the company founded by person X?" requires identifying the company, locating its headquarters, and identifying the country. A model might answer each smaller question correctly in isolation yet give an incompatible answer to the combined question.

We want to investigate whether such failures occur disproportionately around factual errors and whether they provide information that simpler uncertainty measures miss.

This extends the verification problem explored in *Think Before You Trust* [1]. That project models dependent claims as a tree or directed acyclic graph and compares strategies for allocating a limited verification budget. Its framework emphasizes confidence and the potential downstream damage of an error.

Our proposed extension adds a semantic question:

> Do alternative, meaningfully equivalent ways of computing an answer agree?

If this helps identify risky regions of a reasoning graph, it could improve verification decisions.

The project concerns observable answers and structured intermediate claims. It does not assume that a model's written explanation faithfully exposes its internal reasoning.

2 Prior Research and Proposed Contribution

The project builds on several closely related research directions.

Prior work	**Existing contribution**	**Proposed extension**
González and Nori, *Beyond Words* (2023 preprint) [2]	Uses commutative diagrams to formalize hallucination, alignment, and relationships between task execution and model computation.	An operational diagnostic that compares model-generated routes without requiring the correct answer during detection.
Press et al., *Measuring and Narrowing the Compositionality Gap* (2023) [3]	Measures cases where models answer the component questions correctly but fail the combined question.	Tests whether observable composition failures predict errors on individual examples and improve verification allocation.
Chen et al., *Two Failures of Self-Consistency* (2024) [4]	Tests whether replacing intermediate subproblems with the model's own answers preserves its final answer.	Extends composition tests to overlapping regions of a task graph and evaluates their value for factual error detection.
Chytas et al., *Understanding Multi-compositional Learning via Category Theory* (2024) [5]	Develops categorical constraints for concept composition and examines their satisfaction in model representations.	Studies factual question answering and verification decisions using observable outputs.
Yang et al., *Hallucination Detection with Metamorphic Relations* (2025) [6]	Detects hallucinations through violations of expected relationships under controlled semantic mutations.	Tests composition-specific transformations and compares their usefulness against this broader consistency approach.

The general connection between composition, consistency, and hallucination already has substantial precedent. Our potential contribution is therefore narrower:

> A method for measuring local composition failures, evaluating their relationship to factual errors, and using them to allocate verification effort under realistic cost constraints.

Whether this exact combination is novel will require a more exhaustive literature review before making a publication claim.

3 Hypotheses

We propose three hypotheses.

H1. Predictive value. Greater compositional inconsistency is associated with a higher probability of factual error, even after accounting for task difficulty, path length, confidence, and ordinary answer disagreement.

H2. Localization. Overlapping composition tests identify regions of a reasoning graph containing errors more effectively than confidence or graph structure alone.

H3. Practical value. A verification policy using composition information detects more false claims, or leaves fewer undetected false claims, at the same total cost as competing policies.

The principal alternative explanation is that composition tests merely reproduce familiar uncertainty signals. A model may disagree across routes because it is uncertain, because the prompts differ, or because longer tasks are harder.

The experiment must therefore establish whether composition-specific comparisons add useful information beyond those effects.

4 Categorical Formulation

For the initial experiment, we will restrict attention to tasks that can be represented as chains of typed operations with sufficiently unambiguous answers.

For example, consider

$$f : \text{Person} \to \text{Company}, \qquad g : \text{Company} \to \text{Country}.$$

Here, f identifies a particular company founded by a person, while g identifies the country containing that company's headquarters. The dataset must specify enough context for these operations to be well-defined.

We compare two implementations:

- **Direct route:** Ask the model to perform the composite operation $g \circ f$.
- **Stepwise route:** Ask the model to perform f, then apply g to its generated answer.

For a deterministic implementation, preservation of composition would require

$$\widehat{F}(g \circ f)(x) = \widehat{F}(g)(\widehat{F}(f)(x)).$$

The notation $\widehat{F}$ denotes a candidate implementation of the task structure. We do not assume it is a functor; preservation of composition is precisely what we are testing.

Because model outputs are stochastic, our operational score will compare answer distributions:

$$\delta_{f,g}(x) = d(P_{\text{direct}}(\cdot \mid x), P_{\text{stepwise}}(\cdot \mid x)).$$

A candidate distance is total variation distance over normalized answer identities:

$$d_{\text{TV}}(P, Q) = \frac{1}{2} \sum_{a \in \mathcal{A}} |P(a) - Q(a)|,$$

where $\mathcal{A}$ is the normalized answer space, including explicit outcomes for abstention and invalid responses.

The exact estimator and sampling procedure will be selected during development and frozen before testing.

This formulation imposes an important requirement: the two routes must represent equivalent tasks. Sharing a starting point and endpoint does not, by itself, make two paths equivalent. The categorical structure specifies which compositions and path equalities are justified.

5 Data Requirements

Each example should contain, or allow us to construct, the following information.

Required information	**Purpose**
Original multi-hop question	Defines the overall task.
Component questions and dependency structure	Identifies operations and valid composition tests.
Independently checked intermediate answers	Supports evaluation of errors at individual steps.
Independently checked final answer	Supports evaluation of overall factual correctness.
Supporting documents or factual evidence	Allows independent verification and adjudication.
Answer aliases or entity identifiers	Prevents wording differences from being counted as factual disagreement.
Model outputs, prompts, sampling settings, and token usage	Supports reproducibility and cost comparisons.

Ground-truth answers and correctness labels will be withheld from the detector. It may access the question, a question-only decomposition, the model's generated answers, and the evidence permitted by the experimental condition.

Any decomposition derived from dataset annotations must have answer-bearing information removed. All comparison methods will receive the same permitted structural information.

The initial dataset candidates are:

- **MuSiQue** [7]: A strong starting point because its questions are constructed by composing smaller questions, with examples requiring two to four hops.
- **2WikiMultiHopQA** [8]: A second dataset designed to support evaluation of reasoning steps using structured and unstructured information.
- **HotpotQA** [9]: A possible later robustness test with sentence-level supporting facts. Additional annotation would be needed wherever intermediate answers are not directly available.

Initially, we will select the subset of questions with clear chain structures and unambiguous relations. More general branching graphs can follow after the basic test is established.

A practical starting scale is approximately **100 pilot questions**, followed by a larger development and held-out evaluation set. The final sample size will depend on observed error frequency, uncertainty in the estimated effect, and available compute.

6 Experiment Design

6.1 Stage A: Construct and Audit Task Representations

For each question, identify the component operations, their dependencies, and the direct-versus-stepwise comparisons that should be equivalent.

Manually audit a sample of these representations. Exclude or separately categorize ambiguous relations, changing facts, and invalid decompositions using rules established before examining detector performance.

Split data by original question and, where feasible, by recurring entities or question templates. All probes and paraphrases derived from one question must remain in the same split.

6.2 Stage B: Generate Original Answers and Intermediate Claims

Ask the model to answer the question and produce a concise, structured record of intermediate answers. Collect these outputs before running diagnostic probes.

Use the same original outputs for every verification policy. This prevents different policies from receiving easier or harder initial answers.

Begin with one model during the pilot, then evaluate at least one additional model to test whether the signal generalizes.

The primary condition will use a fixed document context containing task evidence and distractors. Direct and stepwise routes will receive the same permitted evidence. A separate closed-book condition can test whether findings extend to questions answered from model knowledge.

6.3 Stage C: Run Composition Tests Without Correctness Labels

For each eligible two-step region:

1. Fix a common input using the question or the model's own upstream answer.
2. Sample answers to the composite question directly.
3. Independently sample the first component, then feed its generated answer into the second component.
4. Normalize the resulting endpoint answers.
5. Measure the discrepancy between the direct and stepwise distributions.

The stepwise route must use model-generated intermediate answers, never substituted gold answers.

Use fresh contexts for independent probes to reduce anchoring. Preserve invalid outputs and abstentions as explicit outcomes.

To distinguish composition failure from sampling noise, also compare independent samples generated through the **same route**. Repeatedly sampling one prompt provides a necessary control for ordinary answer instability.

For a chain of several operations, apply overlapping two-step tests. A failed test marks its participating region as suspicious; it does not establish which individual claim is false.

6.4 Stage D: Independently Label Factual Errors

Evaluate the original intermediate claims and final answers using reference answers and supporting evidence.

Keep separate labels for:

- False factual claims.
- Unsupported claims whose truth cannot be established.
- Invalid inference steps.
- Correct claims.
- Abstentions or unresolved outputs.

This distinction matters when a model follows a wrong premise consistently. A subsequent inference may be valid conditional on that premise while still producing a factually incorrect claim.

Generated claims about entities outside the reference answer path may require additional evidence or manual adjudication. Missing evidence alone must not automatically count as falsity.

6.5 Stage E: Test Predictive Value and Localization

Compare a baseline error predictor using confidence, task length, relation type, question length, and ordinary disagreement against the same predictor with composition scores added.

Any learned predictor will be fitted on development data only. Difficulty proxies must be available without test correctness labels; independently measured difficulty may be used for a separately identified explanatory analysis.

Primary evaluations will include:

- Precision–recall performance for factual error detection.
- Improvement over the baseline predictor on held-out questions.
- Precision among the highest-ranked suspicious regions.
- Fraction of erroneous claims covered by selected regions.

Use paired comparisons and confidence intervals clustered by original question. Multiple probes from one question are related observations, not independent examples.

6.6 Stage F: Compare Verification Policies Under Equal Budgets

Compare the following policies.

Policy	**Selection rule**
Random	Select claims uniformly.
Confidence	Prioritize low-confidence claims.
Structural	Prioritize claims with substantial downstream influence.
Existing dependency-aware policy	Combine estimated error risk and downstream damage using the repository's approach.
Ordinary consistency	Prioritize claims with unstable repeated answers.
Composition	Prioritize regions with high composition discrepancy.
Combined policy	Combine composition information with confidence and downstream influence.

Any rule that converts a suspicious region into a claim selection must be fixed before testing. For example, begin by checking the region's endpoint, then inspect its intermediate claim if uncertainty remains.

Every policy will use the same factual verifier and permitted evidence.

Diagnostic calls count against the budget. With three samples per route, a two-step composition test requires three direct calls and six component calls: nine additional calls before normalization or verification costs.

We will report token usage, model calls, and latency. The primary budget comparison will charge each method for all diagnostic and verification work, allowing cheaper methods to spend their savings on additional checks.

First evaluate verification and flagging alone. A later repair experiment can test whether correcting selected claims and regenerating dependent answers improves final accuracy.

7 Controls, Limitations, and Interpretation

Several outcomes would limit the hypothesis.

A model may give the same wrong answer through every route. Such errors will evade a consistency-based detector. Conversely, disagreement may arise from ambiguity or prompt sensitivity even when the original answer is correct.

Recent work on correctness, consistency, and prompt multiplicity emphasizes the need to distinguish correctness from consistency when evaluating hallucination detection [10]. That distinction is central to this proposal.

We will therefore include the following controls:

- Compare composition tests with repeated-answer disagreement at matched cost.
- Repeat selected tests using alternative, equivalent prompt templates.
- Evaluate whether results persist within comparable path lengths and relation types.
- Audit normalization errors and ambiguous questions.
- Report both consistently wrong answers and inconsistent but correct original answers.
- Compare against an established transformation-based detector such as MetaQA [6].

The primary study establishes predictive association and practical utility. It cannot, by itself, establish that compositional inconsistency causes hallucination.

8 Expected Outcomes and Project Outline

A positive result would show that composition tests provide useful information beyond existing uncertainty measures and improve verification outcomes after accounting for their cost.

A partial result might show improved error prediction without improved verification efficiency. This would motivate cheaper or selectively applied probes.

A negative result would also be informative: composition failures might largely reflect ordinary uncertainty, occur too far from the original error to guide verification, or cost more than the checks they replace.

The project can proceed through five milestones:

1. **Formalization:** Define the supported task fragment, valid path equivalences, and composition score.
2. **Pilot:** Audit roughly 100 questions and measure feasibility, error prevalence, and probe cost.
3. **Detection study:** Freeze the protocol and evaluate predictive value on held-out questions.

4. **Verification study:** Integrate the signal into the existing graph framework and compare policies under equal budgets.
5. **Robustness and write-up:** Evaluate another model and dataset, characterize failure cases, and release the protocol and evaluation code.

The intended deliverables are a task representation specification, a labeled evaluation set, reproducible composition probes, comparisons against existing detectors, and an analysis of verification performance versus total cost.

The research claim we aim to test is:

> **Local failures to preserve task composition provide information about factual errors that can improve the allocation of limited verification effort.**

References

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